Spatially and Relationally Aware Image Editing via a Multi-Model Pipeline

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Abstract

Current end-to-end computer vision models, particularly diffusion models, struggle with precise spatial reasoning and relational editing. Because standard diffusion relies on latent space pattern matching rather than an inherent understanding of physical constraints or 3D spatial geometry, these models frequently hallucinate new features or resort to simple pixel-blending rather than accurate physical manipulation. To address this, we propose a modular "mixture of experts" pipeline that separates semantic reasoning from deterministic spatial mathematics and final pixel rendering. Leveraging a Vision-Language Model (VLM) as an orchestrator, the system extracts structured spatial intents and passes them to a suite of specialized perception models (Grounding DINO, CLIP, SAM) for pixel-perfect segmentation. A deterministic spatial reasoning engine calculates 3D depth, 2D affine transformations, and bounding masks, which guide a localized inpainting diffusion model to execute photorealistic, geometrically sound edits.

1. Introduction

Achieving accurate visual question answering (VQA) and precise, geometry-aware image editing requires a deep understanding of both visual appearance and the contextual semantics that control object interactions. Traditionally, tasks requiring complex spatial reasoning rely on intermediate representations, such as scene graphs, to distill salient pairwise relationships between objects. While recent frameworks have successfully leveraged Vision Language Models (VLM) for open-world spatial reasoning without dataset-specific fine-tuning, these approaches are inherently analytical and do not extend seamlessly into image modification. To bridge this gap, we aim to develop an interactive, unified multimodal framework capable of performing both VQA and controllable image editing. Given an input image and a natural language prompt, the system allows the user to either perform a VQA-style spatial reasoning task or generate an edited image that applies a requested structural modification, such as moving or adding objects under strict spatial constraints.

A key challenge in developing structured editing frameworks is translating semantic understanding into actionable pixel-level changes. Initially, our preliminary investigations explored the generation of exhaustive Scene Graphs to map spatial relations. However, we found that translating these graphs back into text to guide generative models consistently increased hallucinations and noise. Ultimately, there is no shared embedding space between a structured scene graph and images. Consequently, our final framework pivots away from scene graphs entirely. Instead, we directly utilize the native reasoning capabilities of a VLM to parse user prompts into deterministic JSON instructions, bridging the gap between semantic intent and spatial execution.

While deep learning foundation models are highly capable, we integrate scene graphs to explore a hybrid approach using elements of Symbolic AI. Purely neural approaches can obscure spatial reasoning, whereas symbolic representations offer enhanced explainability, improved trust in complex logical reasoning, and zero-shot data efficiency. Despite drawbacks like information bottlenecks, brittleness, and optimization challenges , scene graphs provide a uniquely transparent way to explore how Vision-Language Models understand reality, which is highly practical given our limited time and compute resources.

2. Related Work

2.1. Scene Graphs for Visual Reasoning

VQA and generative image editing have historically struggled with compositional reasoning and precise spatial control. Standard VQA systems often fail on nuanced visual arrangements, while text-to-image diffusion models alter global layouts indiscriminately during localized edits. Scene Graph Generation (SGG) addresses this by recognizing objects and their pairwise relationships, benefiting downstream tasks such as VQA and image editing. Frameworks like SGEdit and VENUS use LLMs to translate user instructions into structured scene graph modifica-

tions, while models like SEELE leverage these graphs for accurate subject repositioning. [2] [10] [9] [6] [8]

2.2. The Era of Zero-Shot Vision-Language Models

Most SGG methods rely on dataset-specific supervision, limiting their use in open-world settings. Vision-Language Models (VLMs) have enabled a paradigm shift, allowing frameworks like OwSGG to treat SGG as a zero-shot structured-reasoning problem. However, since VLMs struggle with spatial localization and often hallucinate implausible interactions, state-of-the-art approaches supplement them with auxiliary modules such as open-vocabulary detectors and monocular depth estimators, motivating our pipeline design.

2.3. Open-World Perception and Grounding DINO

Grounding DINO[3] enables zero-shot detection of arbitrary text-defined object categories by integrating transformer-based detection with vision-language pretraining, generalizing to unseen classes without retraining. In our pipeline, it localizes object instances via bounding boxes, which are then passed to Segment Anything (SAM)[5] to produce high-quality instance masks, together enabling open-vocabulary instance segmentation without large-scale retraining.

2.4. Monocular Depth Estimation

Accurate spatial reasoning requires geometric context to filter implausible object relationships. Depth estimation has evolved from supervised models to scalable foundation approaches, with Depth Anything V2 achieving strong generalization and sharp boundary prediction through a synthetic-to-real teacher-student framework with large-scale pseudo-labeling. [11]

2.5. Generative Inpainting and Harmonization

Precise image editing requires robust background restoration and seamless object integration. Large Mask Inpainting (LaMa) [7] provides a strong baseline for efficient, structure-aware object erasure. Meanwhile, latent diffusion models like Stable Diffusion XL (SDXL) [4] offer great fidelity for localized texture generation and boundary harmonization. Because purely neural inpainting can introduce lighting inconsistencies, classic techniques like Poisson image blending remain highly relevant. Poisson blending matches local illumination and color while strictly preserving the pasted object's internal structural integrity.

2.6. Application to our Pipeline

The methods surveyed above directly motivate our final four-stage architecture. While traditional visual reasoning relies on Scene Graphs (Section 2.1), our preliminary tests revealed these introduce information bottlenecks. Instead, we leverage the zero-shot reasoning of modern VLMs to act as a direct orchestrator, outputting structured JSON intents. To ground these intents, we utilize the Grounded SAM paradigm (Section 2.3) for pixel-precise instance masks, while Depth Anything V2 (Section 2.4) provides the geometric context needed for mathematical spatial transforms. Finally, state-of-the-art inpainting models like LaMa and SDXL (Section 2.5) execute the localized background repair and edge harmonization.

3. Preliminary Investigations

Before building the final "mixture-of-experts" system, we ran a preliminary phase to check the two most failure prone parts of the problem: (1) controlled object level editing and (2) structured spatial understanding. The key motivation was that end to end diffusion models are highly photorealistic but often unreliable for precise spatial moves and hard to debug. A modular approach makes failures traceable to a specific component (grounding, masking, geometry, or inpainting).

3.1. Preliminary Pipeline 1: Modular, Zero-Shot Image Editing

The purpose of this was to confirm that we could do a clean object extraction and reinsertion without relying on a single generative model to "do everything."

- **Detection + segmentation :** We locate the object and then generate a pixel accurate binary mask so that edits affect only the intended subject. In the final direction of the project, this idea is preserved as GroundingDINO $\rightarrow$ CLIP $\rightarrow$ SAM (high recall candidates, then select best match, then segment)

- **Background repair (inpainting):** After removing an object, we need to heal the background using inpainting. This is done either with LaMa (fast erasure) or SDXL inpainting (higher quality but slower), depending on the approach taken.

Diffusion is most reliable when restricted to local regions (removal area or boundary seam), rather than regenerating large parts of the image. This influenced our later design where SDXL is used mainly for edge harmonization on a thin boundary mask

3.2. Preliminary Pipeline 2: Scene Graph Generation with VLMs

Our second exploratory pipeline explored zero-shot Open-World Scene Graph Generation, inspired by a paper by Dutta et al [1]. The process begins with entity generation, where a VLM creates a free-form list of potential objects, which an embedding-based matching mechanism

(SimCSE) then aligns to known dataset categories. Grounding DINO localizes these entities with bounding boxes (Figure 1a), filtering out non-physical hallucinations. To avoid the computational expense of evaluating every possible object pair, we fuse semantic VLM scores with 3D spatial data from a Depth-Anything-V2 monocular depth map (Figure 1b) to isolate the top-k most probable interacting pairs (Figure 1c). Finally, the VLM predicts specific relational predicates for these refined pairs, mapping them into an instance-aware scene graph of structured subject-predicate-object triplets (Figure 1d).

We validated this framework on the VG150 dataset and custom vocabularies. The pipeline successfully mapped free-form predictions to approved labels (e.g., "napkin" to "paper"), localized objects, and extracted coherent spatial relationships (e.g., a pizza inside a box) across distinct scenes. However, translating these generated scene graphs back into text to guide generative image editing models proved ineffective, often increasing hallucinations compared to using the VLM's native reasoning and spatial understanding. Consequently, our final system abandons intermediate scene graphs, instead using the VLM as an orchestrator to directly output actionable JSON intents for deterministic spatial editing.

4. Primary Methodology

To overcome the spatial reasoning limitations of end-to-end diffusion models, our final framework employs a modular "mixture of experts" pipeline that separates semantic reasoning from deterministic spatial mathematics and final pixel rendering. This architecture processes an input image and a natural language instruction through four sequential stages. First, a Vision-Language Model acts as an orchestrator to extract structured intents. Next, specialized perception models isolate the target objects. A deterministic spatial reasoning engine then calculates the new geometric configuration, which finally guides a localized generative model to execute a photorealistic, physically sound edit.

4.1. Stage 1: Intent Extraction

The pipeline is initialized by a large Vision-Language Model (VLM), specifically Gemma-4-E4B, which acts as the orchestrator. The VLM ingests the original image and the natural language edit instruction, serving as the primary reasoning engine. It strictly parses the instruction into a structured JSON dictionary containing the task type (e.g., "move") and the target interaction operator (e.g., "center"). Crucially, for both the subject and reference objects, the VLM generates a concise noun label optimized for object detection, alongside a highly detailed visual description optimized for downstream similarity matching.

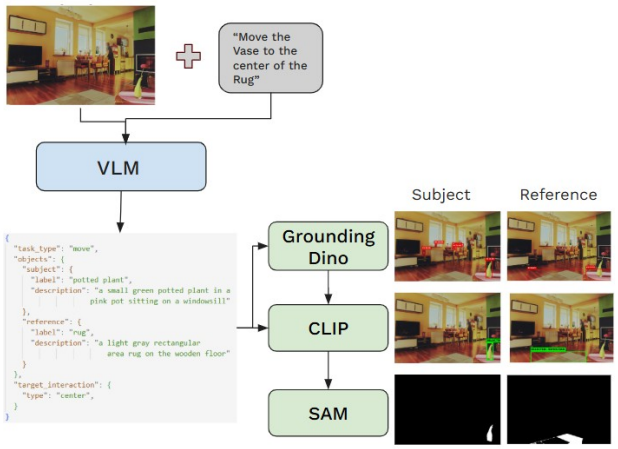

Figure 1. Grounding and Masking Example

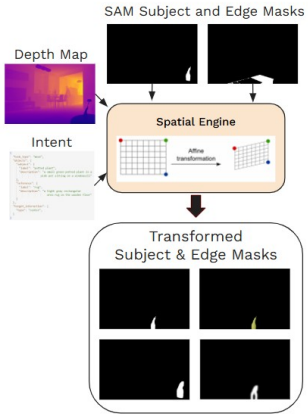

Figure 2. Spatial Editing Example

4.2. Stage 2: Cascading Grounding and Masking

Specialized, single-task vision models execute the extraction of scene elements based on the VLM's structured outputs, as illustrated in Figure 1. The process begins with candidate generation, where Grounding DINO utilizes the VLM's short noun labels to locate multiple candidate bounding boxes within the scene. To ensure maximum selection precision, CLIP then performs similarity filtering by evaluating all of these DINO-generated candidates against the VLM's detailed visual descriptions and isolating the highest-scoring box. Finally, the Segment Anything Model (SAM) takes this winning bounding box and generates a pixel-perfect binary mask of the target instance.

4.3. Stage 3: Spatial Reasoning Engine

This programmatic intermediate layer mathematically calculates the new spatial configuration, as demonstrated in Figure 2. The engine utilizes Depth Anything V2 to generate a monocular depth map, which is used to calculate a re-

alistic scale factor between the subject and reference planes. Based on the requested relational operator, the engine computes 2D affine transformations to translate and warp the extracted RGB pixels. Finally, it programmatically dilates and erodes the transformed mask to isolate an "Edge Mask" for localized blending.

4.4. Stage 4: Generative Execution

The final stage executes the physical manipulation. We evaluate three variants of the generative execution module that differ in how the source region is removed, how the moved subject is integrated, and how local boundary refinement is applied.

4.4.1. Approach 1: LaMa removal + shadow + Poisson blending + SDXL edge-only refinement

This approach first uses LaMa inpainting to erase the subject from its original location using the dilated removal mask. After aligning masks and the warped subject to the original resolution, we generate a shadow mask from the transformed object mask and apply it to the erased background to improve physical realism. The warped subject is then integrated using Poisson blending to match local illumination and color. Finally, Stable Diffusion XL is applied only on a thin background-only edge mask, namely an edge band around the pasted object that excludes the object pixels, to harmonize boundary transitions without changing the object appearance. This approach follows the key design principle of keeping diffusion localized in order to avoid global hallucinations.

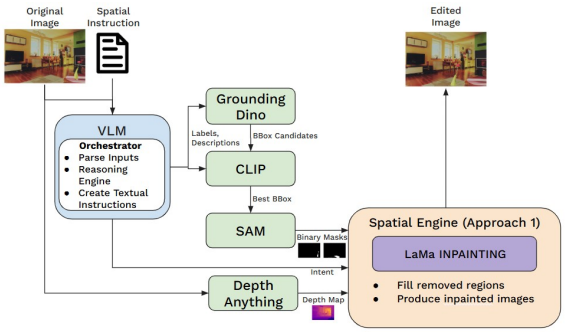

Figure 3. Approach 1 pipeline visualization

4.4.2. Approach 2: SDXL-only removal + hard paste + SDXL edge blending

In this variant, SDXL inpainting replaces LaMa for the initial removal step, generating a clean background directly from the original image and the removal mask. After that, the warped subject is hard-composited onto the background using the transformed mask. As in Approach 1, SDXL is then optionally used for edge blending only using a background-only edge mask that avoids modifying the object itself. This approach can produce high-quality removal, but it is typically slower and may introduce stronger generative variation because SDXL is responsible for larger inpainted regions.

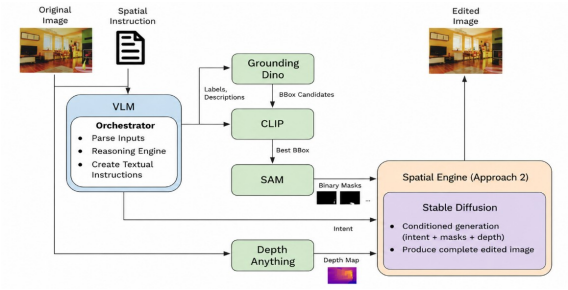

Figure 4. Approach 2 pipeline visualization

4.4.3. Approach 3: LaMa removal → SDXL refinement of removal area → shadow → hard paste → Poisson → SDXL edge refinement + brightness sanity check

This approach combines the strengths of both variants. LaMa provides a fast initial removal, and SDXL then refines only the removal region to reduce LaMa artifacts and improve texture continuity. Next, we apply shadowing, per-

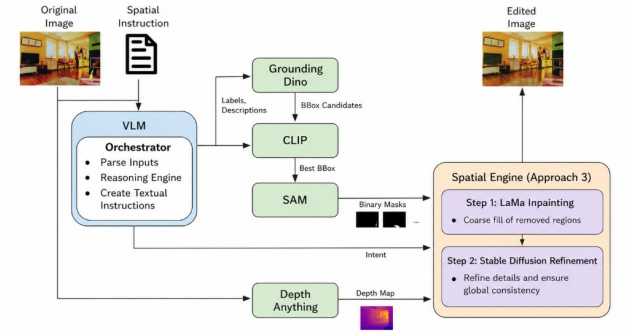

Figure 5. Approach 3 pipeline visualization

form a hard paste, and then use Poisson blending for color matching. We again run SDXL only on background edges for seamless boundaries, and finally lock object pixels via compositing to preserve the moved subject. A brightness sanity check is included to detect failure cases, such as overly dark or overly bright outputs, and to fall back to the hard-paste result when needed. In our presentation discussion, this hybrid strategy is reported as slightly better overall in image quality than the SDXL-only approach.

4.5. Dataset

To evaluate spatially and relationally aware image editing, we use two complementary data sources: COCO for nat-

ural, unconstrained scenes and a synthetic CLEVR-based dataset for controlled, metric-driven testing.

Common Objects in Context (COCO): COCO provides complex everyday scenes with multiple objects, clutter, and diverse indoor/outdoor contexts.This is important as our pipeline depends on correctly grounding both a subject and a reference object before performing a spatial edit. We used COCO images directly as end to end test inputs. Images are fetched by URL and passed through the 4 stages of the pipeline. This makes COCO a practical benchmark for testing robustness under real world ambiguity.

Synthetic dataset (CLEVR + VLM-generated commands): To obtain repeatable ground truth style evaluation cases, we also generate a synthetic dataset using CLEVR, which provides known object properties, explicit spatial relationships, and 3D coordinates. A VLM (Gemma 4) generates casual user prompts and structured commands for object movement, producing evaluation metadata such as structured ground truth commands, projected destination coordinates, and required scale factors.

In total, we created 150 images with commands and ground truth tests to support systematic spatial accuracy analysis. This synthetic dataset complements COCO by enabling controlled measurement when real-image ground truth is unavailable.

5. Evaluations & Discussion

Tests were ran in Google Colab using an A100 GPU. Dataset Generation and Image Editing took around 90 minutes each. We used the chain of thought LLM / VLM prompting for both generation and evaluation. The largest bottleneck was VLM Inference. Table 1 shows the models we used for implementation.

Model	Implementation
DINO	Grounding Dino Base
CLIP	CLIP ViT Base Patch32
SAM	SAM ViT Huge
Depth	Depth Anything V2 Metric Indoor Base
Diffusion	Stable Diffusion XL 1.0 Inpainting
VLM	Gemma 4 E4B (it)

Table 1. Models Used in Pipeline

Our final experiments evaluate the end to end system across the synthetic CLEVR-based benchmark, measuring: (1) intent extraction correctness, (2) grounding quality for subject/reference, (3) spatial transform accuracy, and (4) final image preservation/realism.

5.1. Primary Evaluation

To evaluate the pipeline, we perform qualitative tests on COCO images by running a small set of real images via URL with directional prompts, such as ”move left of” or ”place above,” to verify end-to-end behavior under clutter and ambiguity. Furthermore, we conduct CLEVR quantitative tests by running the full pipeline on our synthetic dataset of 150 images, which provides structured commands and spatial ground truth like projected destination coordinates and expected scale factors. The evaluation harness logs outputs and metrics per image and produces an aggregated report. Our results are displayed as the **Hybrid** column of Table 2.

Measuring performance across the pipeline stages reveals the power of the VLM as a ”Director”. The Stage 1 Orchestrator achieved an impressive 99.3% intent accuracy. This demonstrates that large vision-language models are highly effective at parsing natural language and generating structured JSON actions, excelling in both synthetic baseline generation and orchestration.

However, the perception alignment between the VLM, Grounding DINO, and CLIP introduces a significant bottleneck. For the Stage 2 Spatial Experts, ground-truth grounding achieved an accuracy of only 71.3%. These specialized models frequently struggle to find the correct bounding box when faced with full-image ambiguity. A proposed solution for future iterations is to empower the VLM to deterministically crop or alter search areas before passing them to the object detection modules.

Evaluating Stage 3 Spatial Reasoning highlights the distinct trade-offs of using deterministic logic. The deterministic mathematical engine lacks absolute precision and occasionally struggles with scaling, resulting in an average pixel error of 51.8 pixels and an average scale MAE of 14.982. Furthermore, deterministic constraints currently limit operations primarily to simple translations like ”move.” Because of this, pipeline cannot handle complex operations such as rotations, removals, or swaps. Despite these limitations, this deterministic approach provides exceptional structural preservation, eliminating global hallucinations entirely.

To comprehensively evaluate the final generative execution, we measured background preservation and perceptual consistency. The pipeline achieved a background SSIM of 0.976 and a background PSNR of 29.29 dB, indicating that our localized editing approach successfully preserves the original image structure. Finally, with a subject LPIPS of 0.162, the moved object and the resulting edited image retain high visual quality without suffering from unintended perceptual degradation.

5.2. Stage 4 Implementation Ablation Study

To further optimize the final generative execution, we compared three different implementations for Stage 4, as de-

Category	Metric	Hybrid	SD-Only
Orchestrator	Intent Accuracy (↑)	99.3%	98.0%
Spatial Experts	GT Grounding (↑)	71.3%	71.3%
Spatial Reasoning	Avg Pixel Error (↓)	51.8 px	51.4 px
	Avg Scale MAE (↓)	14.982	14.985
Generative Exec.	Background PSNR (↑)	29.29 dB	28.62 dB
	Background SSIM (↑)	0.976	0.974
	Subject LPIPS (↓)	0.162	0.159
Global Baseline	Global PSNR (↑)	27.16 dB	
	Global SSIM (↑)	0.962	
	Global LPIPS (↓)	0.199	

Table 2. Our quantitative evaluation establishes the independent performance metrics for each stage of our overarching architecture. Additionally, it functions as an ablation study to compare our hybrid generative execution strategy against an SD-Only approach.

scribed in Subsection 4.4. We observed that, overall, the hybrid approach performed slightly better in terms of final image quality over the Stable Diffusion only approach. Details of the results on the Stable Diffusion only model are given on the rightmost column of Table 2. Note that we do not report parameters of the first tested approach, as basic qualitative assessment proved it inferior to the other designs.

5.3. Diffusion Model Ablation Study

In a subsequent experiment, we quantitatively evaluated three specific Stable Diffusion pipelines, Stable Diffusion 1.5, Stable Diffusion XL Base, and Stable Diffusion XL Inpainting, using spatial accuracy, background LPIPS, and CLIP-IQA realism as objective metrics. Results are produced in Table 3. All three models achieved a spatial accuracy of 100%, indicating that each pipeline consistently preserved the intended geometric structure of the generated content. The background LPIPS scores were 0.0086 for Stable Diffusion 1.5, 0.0107 for Stable Diffusion XL Base, and 0.0086 for Stable Diffusion XL Inpainting, where lower values correspond to higher perceptual similarity between the synthesized and reference backgrounds. Finally, the CLIP-IQA realism scores were 0.2850 for Stable Diffusion 1.5, 0.2576 for Stable Diffusion XL Base, and 0.2815 for Stable Diffusion XL Inpainting, suggesting that Stable Diffusion XL Inpainting attains a favorable balance between background fidelity and perceptual realism, and that both Stable Diffusion 1.5 and Stable Diffusion XL Inpainting produce more visually realistic outputs than the Stable Diffusion XL Base model under the chosen evaluation protocol.

Ultimately, the overall results indicate that Stable Diffusion 1.5 is the preferable model for this task when model size and efficiency are taken into account. Stable Diffusion 1.5 matches or surpasses the larger XL variants in all reported quantitative metrics, attaining perfect spatial accuracy, the lowest background LPIPS (tied with XL Inpainting), and a CLIP-IQA realism score that is effectively comparable to the best-performing pipeline. Given that Stable Diffusion 1.5 achieves these results with a substantially smaller model, it offers a more favorable trade-off between computational cost and generation quality, making it the more practical choice for resource-constrained deployments or real-time applications.

Model	Spatial Accuracy	Background LPIPS	CLIP-IQA Realism
SD 1.5	100%	0.0086	0.2850
SD XL Base	100%	0.0107	0.2576
SD XL Inpainting	100%	0.0086	0.2815

Table 3. Comparison of Stable Diffusion pipelines.

6. Future Work

While our multi-model pipeline successfully separates semantic reasoning from spatial mathematics, the deterministic logic layer currently limits operations primarily to simple translations (e.g., “move”). Furthermore, the pipeline’s perception and generative stages face distinct challenges: perception alignment between the VLM and Grounding DINO occasionally struggles with full-image ambiguity, while the final generative execution can suffer from inpainting hallucinations and poor edge blending. To directly address these bottlenecks, future iterations could explore integrating larger, newer foundation models. Improving the modules of this pipeline could significantly enhance final image fidelity and resolve the issue of models finding the wrong target objects.

Beyond component upgrades, we future work can evolve this static, feed-forward pipeline into an iterative agentic framework utilizing a ReAct (Reason + Act) [12] loop. The viability of this structure for high-fidelity image editing has been successfully demonstrated by recent works such as Agent Banana [13]. Rather than passing data linearly, the VLM will act as an autonomous agent empowered to dynamically orchestrate a suite of deterministic visual tools. This architecture would allow the VLM to self-correct by visually evaluating intermediate outputs. For example, verifying if the correct bounding box was generated and looping back to adjust its search parameters before finalizing the edit. Ultimately, we aim to explore if the advancing localization and reasoning capabilities of these state-of-the-art VLMs can eventually replace specialized modules like DINO and CLIP entirely, creating a more streamlined and resilient editing agent.

7. Conclusion

To address the limitations of end-to-end diffusion models in precise spatial reasoning and relational editing, this work introduces a modular ”mixture of experts” pipeline.

By separating semantic reasoning from deterministic spatial mathematics and final pixel rendering, the framework utilizes a Vision-Language Model as an orchestrator alongside specialized perception and generative models. This approach successfully eliminates global hallucinations and ensures high-quality, geometrically sound structural preservation during image manipulation.

Quantitative evaluations demonstrated the pipeline's strong performance, particularly highlighting the orchestrator's 99.3% intent accuracy and the system's ability to maintain excellent background and subject visual fidelity. However, the study also identified key bottlenecks within the system, such as a 71.3% accuracy in grounding due to perception ambiguity, as well as limitations in the deterministic engine that currently restrict edits to simple translations rather than complex operations like rotations or swaps.

Moving forward, future work can transform this pipeline into an iterative agentic framework using a ReAct loop, allowing the VLM to autonomously evaluate intermediate outputs and self-correct. Ultimately, advancing the reasoning and localization capabilities of VLMs may streamline the system further by eventually replacing the need for separate, specialized perception modules.

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8. Appendix

8.1. Qualitative Results

We provide qualitative examples of our pipeline in this section. Looking at our qualitative successes in Figure 6, we see that the pipeline effectively executed complex relational instructions. In these examples, the pipeline correctly identifies the correct reference and target item, and moved the reference item to the correct location. Additionally, the pipeline was able to generate realistic shadowing.

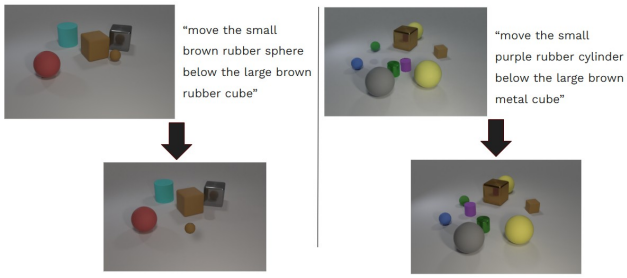

Figure 6. Successful Edits by Pipeline

Our pipeline was unsuccessful in some other ways. A sample of these failure modes is provided in Figure 7. One failure mode, as seen on the left, is perception ambiguity. When instructed to move a small blue metal cylinder, the model instead identified and moved a large blue rubber cylinder. The second common failure mode is poor inpainting and blending. On the right image, when moving the purple metal sphere, the inpainting module hallucinated a cube where the sphere used to be. Also, the reflective surface of the sphere experienced some artifacts in its new location.

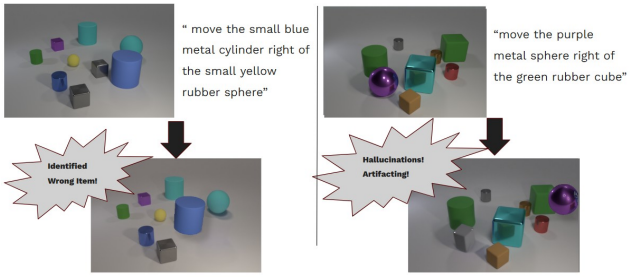

Figure 7. Unsuccessful Edits by Pipeline